

Choice Under (Objective) Uncertainty

Primitives:

$\mathbb{X} = \{x_1, \dots, x_n\}$, a set of outcomes with $n \geq 3$.

$\mathcal{L} = \{p \in \mathbb{R}_+^n \mid \sum p_i = 1\}$ the set of probability distributions (lotteries) on $\mathbb{X}$.

e_i , the element of $\mathcal{L}$ that assigns probability 1 to $x_i \in \mathbb{X}$

$\succsim \subset \mathcal{L} \times \mathcal{L}$ (with $\succ$ and $\sim$ defined in the usual way).

Axioms:

- (A1) $\succsim$ is *complete* and *transitive*.
- (A2) $\succsim$ is (mixture) *continuous*: For any p, q, r in $\mathcal{L}$ with $p \succsim q \succsim r$, there is an α in $[0, 1]$ such that

$$\alpha p + (1 - \alpha)r \sim q.$$

- (A3) $\succsim$ satisfies *independence*: For any p, q, r in $\mathcal{L}$ and $\alpha \in (0, 1)$ we have

$$q \succsim p \text{ if and only if } \alpha q + (1 - \alpha)r \succsim \alpha p + (1 - \alpha)r.$$

Theorem (Expected Utility Representation) The binary relation $\succsim$ on $\mathcal{L}$ satisfies (A1)-(A3) if and only if there is a real-valued function u on $\mathbb{X}$ such that for any p, q in $\mathcal{L}$

$$q \succsim p \text{ if and only if } \sum_{i=1}^n u(x_i)q_i \geq \sum_{i=1}^n u(x_i)p_i. \quad (1)$$

Moreover, if u is replaced by a real-valued function v on $\mathbb{X}$ then (1) remains true if and only if $v = a + bu$ for real numbers a and b , with $b > 0$.

Exercise: Prove the last sentence of the Theorem.

Write $U(p) = \sum u(x_i)p_i$, so $U(e_i) = u(x_i)$. Since $p = \sum p_i e_i$, it follows that $U(\sum p_i e_i) = \sum U(e_i)p_i$, so the representation U of $\succsim$ is linear. The function U on $\mathcal{L}$ is called the *von Neumann-Morgenstern* (vN-M) *utility*. I will often abuse terminology and refer to the function u on $\mathbb{X}$ as the vN-M utility.

Lemma 1 (Monotonicity) Suppose independence (A3) holds and for p, q in $\mathcal{L}$, $q \succ p$. Then

$$\beta q + (1 - \beta)p \succsim \alpha q + (1 - \alpha)p \text{ if and only if } 0 \leq \alpha \leq \beta \leq 1.$$

Proof sketch of Lemma 1 If $1 \geq \lambda^* \geq \lambda \geq 0$ then $\lambda^* q + (1 - \lambda^*)p = (\lambda^* - \lambda)q + (1 - \lambda^* + \lambda)s$ and $\lambda q + (1 - \lambda)p = (\lambda^* - \lambda)p + (1 - \lambda^* + \lambda)s$, where $s = (\lambda q + (1 - \lambda^*)p)/(1 - \lambda^* + \lambda)$. Set $\beta = \lambda^* \geq \alpha = \lambda$ to prove one direction and $\alpha = \lambda^* > \beta = \lambda$ to prove the other.

Lemma 2 (Best and Worst Lotteries) (i) (A1) implies that there are $\succsim$ -best and $\succsim$ -worst degenerate lotteries. (ii) Suppose that (A1) holds, and relabel the sure outcomes so that e_n is a best degenerate lottery and e_1 a worst. If (A3) holds, then $e_n \succsim p \succsim e_1$ for all $p \in \mathcal{L}$.¹

Proof sketch of Lemma 2. (i) I use induction on the size of the outcome set $\mathbb{X}$. If $\mathbb{X}$ has just 1 element then the single element is both best and worst by completeness. Suppose now that any outcome set $\mathbb{X}$ with $k - 1 \geq 1$ elements has $\succsim$ -best and $\succsim$ -worst degenerate lotteries. Consider now an outcome set $\mathbb{X}$ with k elements. Remove the k -th element, x_k , and let $\mathbb{X}_- = \mathbb{X} - \{x_k\}$. By hypotheses there is a $\succsim$ -best degenerate lottery of elements of $\mathbb{X}_-$; relabel the outcomes so that it is e_{k-1} . By completeness, either $e_{k-1} \succsim e_k$ or $e_k \succ e_{k-1}$. If $e_{k-1} \succsim e_k$ then e_{k-1} is a best degenerate lottery of elements of $\mathbb{X}$. If $e_k \succ e_{k-1}$, then $e_k \succ e_{k-1} \succsim e_i$ for $i = 1, \dots, k - 2$, so by transitivity, e_k is the best degenerate lottery of elements of $\mathbb{X}$. The existence of a worst degenerate lottery is similar. (ii) Suppose that (A1) and (A3) hold, and let e_n be the best and e_1 the worst degenerate lottery. To prove (ii) I use induction on the size of the support of $p \in \mathcal{L}$. Obviously if p has one-point support (is degenerate) then $e_n \succsim p \succsim e_1$ by part (i). Suppose that $e_n \succsim q \succsim e_1$ for any $q \in \mathcal{L}$ with support of at most $k - 1$ elements for $k - 1 \geq 1$. Suppose that p has at most k elements in its support. Write $p = (1 - p_i)r + p_i e_i$, where x_i is in the support of p and r is a lottery with at most $(k - 1)$ -point support. Since $e_n \succsim e_i$ and $e_n \succsim q$ (by hypothesis), apply (A3) twice to find that $e_n \succsim p$. The proof that $p \succsim e_1$ is similar. $\square$

Proof of Theorem: The proof that if there is a real-valued function u on $\mathbb{X}$ satisfying (1) then (A1)-(A3) hold I leave as an (easy) exercise. Suppose that (A1)-(A3) hold. By Lemma 2 there are $\succsim$ -best and $\succsim$ -worst lotteries in $\mathcal{L}$ and these are degenerate. Relabel the sure outcomes so that e_n is the best and e_1 the worst element of $\mathcal{L}$. If all the sure elements are indifferent, then by Lemma 2 any two elements of $\mathcal{L}$ are indifferent, and any constant function u satisfies (1). Suppose that there are at least two elements of $\mathcal{L}$ that are not indifferent. By transitivity $e_n \succ e_1$ (you proved this fact in Assignment I). By mixture continuity, for each $p \in \mathcal{L}$, there is a number $U(p) \in [0, 1]$ such that

$$p \sim U(p)e_n + (1 - U(p))e_1 \tag{2}$$

and it is unique by monotonicity (Lemma 1). By transitivity $q \succsim p$ if and only if $U(q)e_n + (1 - U(q))e_1 \succsim U(p)e_n + (1 - U(p))e_1$. And by monotonicity the last holds if and only if $U(q) \geq U(p)$, so $U : \mathcal{L} \rightarrow [0, 1]$ represents $\succsim$. By independence

$$\begin{aligned} \alpha q + (1 - \alpha)p &\sim \alpha[U(q)e_n + (1 - U(q))e_1] + (1 - \alpha)[U(p)e_n + (1 - U(p))e_1] \\ &= [\alpha U(q) + (1 - \alpha)U(p)]e_n + [1 - \alpha U(q) - (1 - \alpha)U(p)]e_1 \end{aligned} \tag{3}$$

But $\alpha q + (1 - \alpha)p \sim U(\alpha q + (1 - \alpha)p)e_n + (1 - U(\alpha q + (1 - \alpha)p))e_1$. By monotonicity and (3), $U(\alpha q + (1 - \alpha)p) = \alpha U(q) + (1 - \alpha)U(p)$. By induction, it follows that, for every $p \in \mathcal{L}$, $U(p) = U(\sum_i p_i e_i) = \sum_i U(e_i) p_i$. Set $u(x_i) = U(e_i)$. $\square$

¹MWG assume directly that there are best and worst lotteries, while Jehle and Reny assume outright the existence of degenerate lotteries e_n and e_1 such that for every $p \in \mathcal{L}$, $\alpha e_n + (1 - \alpha)e_1 \sim p$ for some $\alpha \in [0, 1]$.